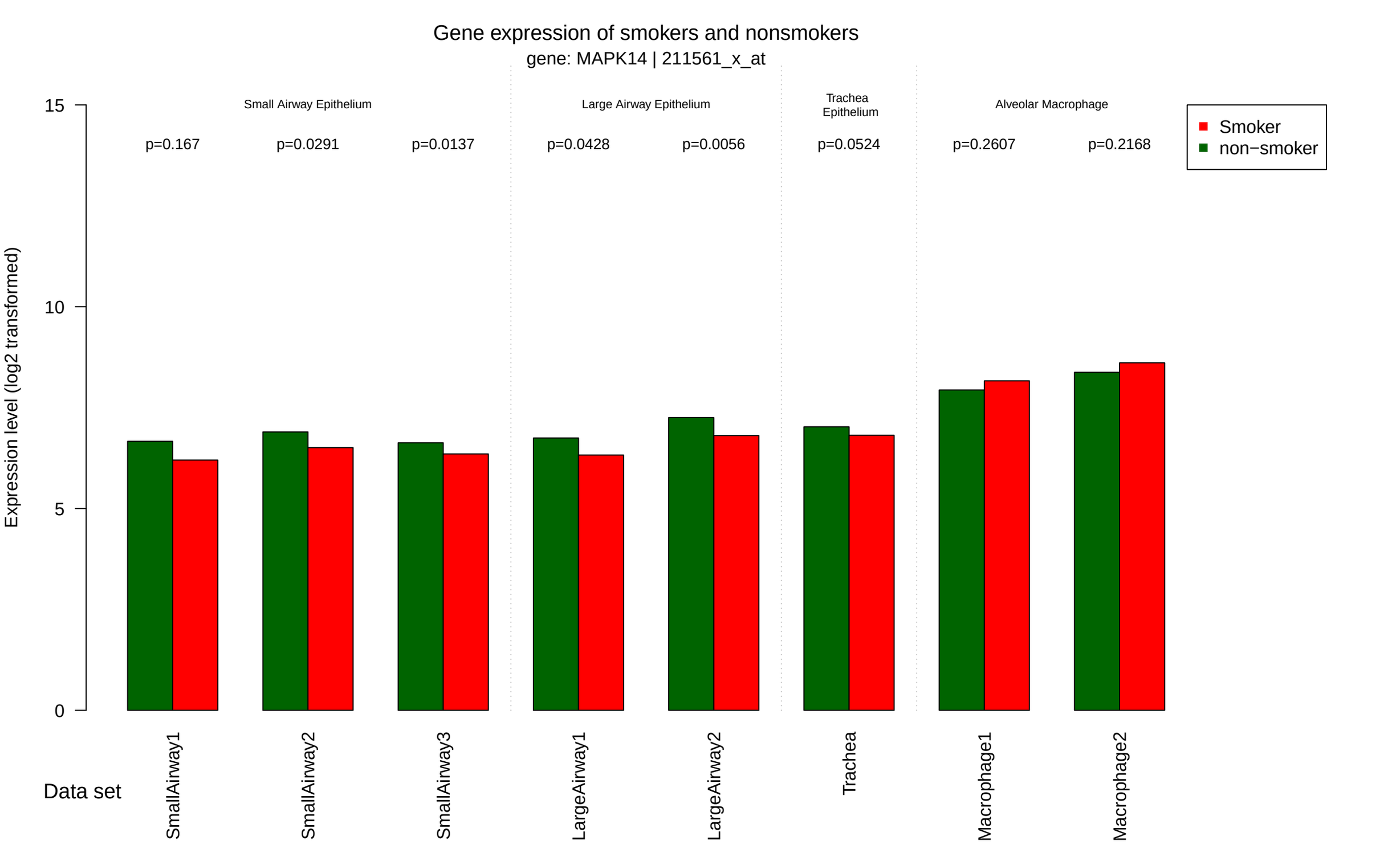
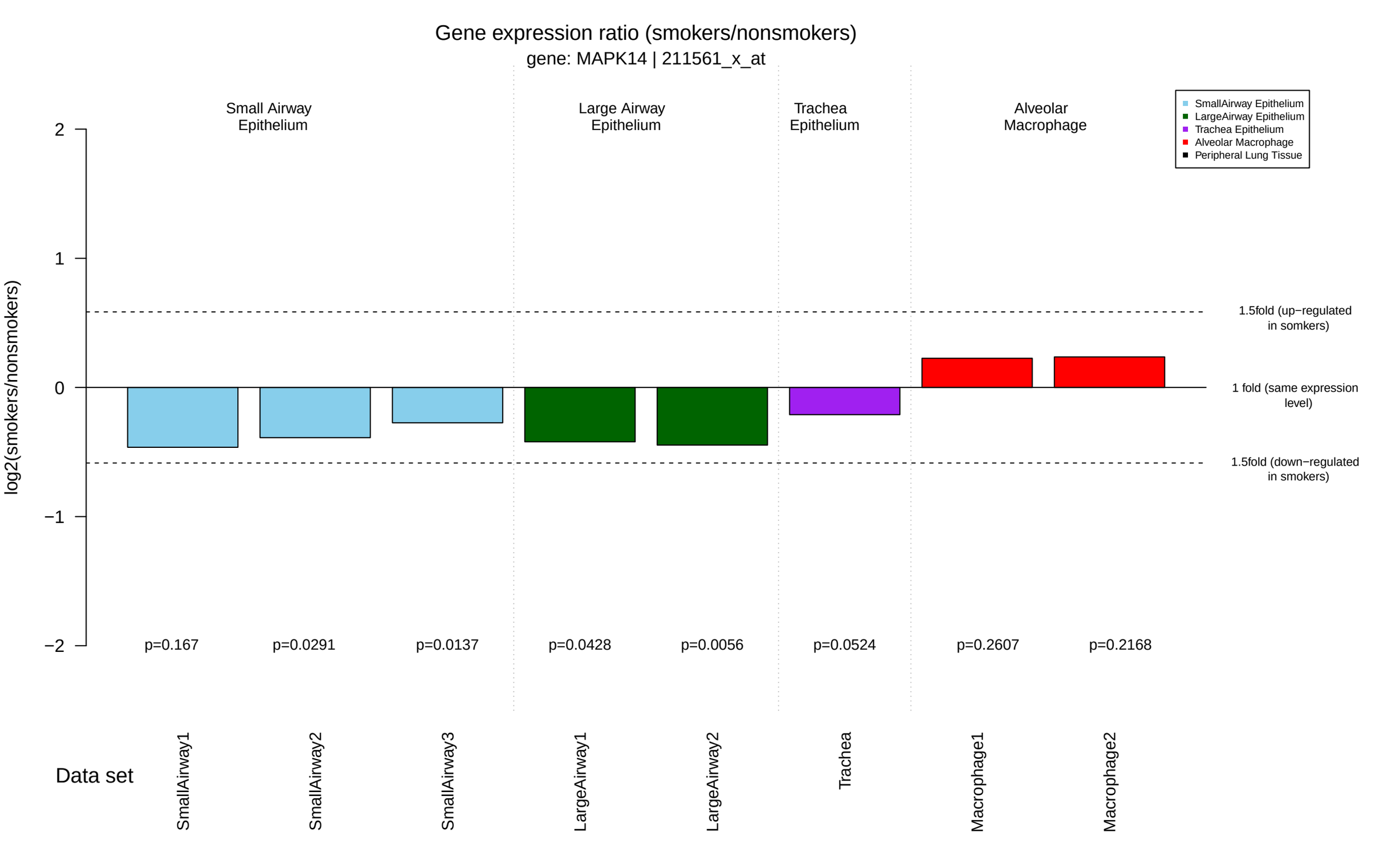
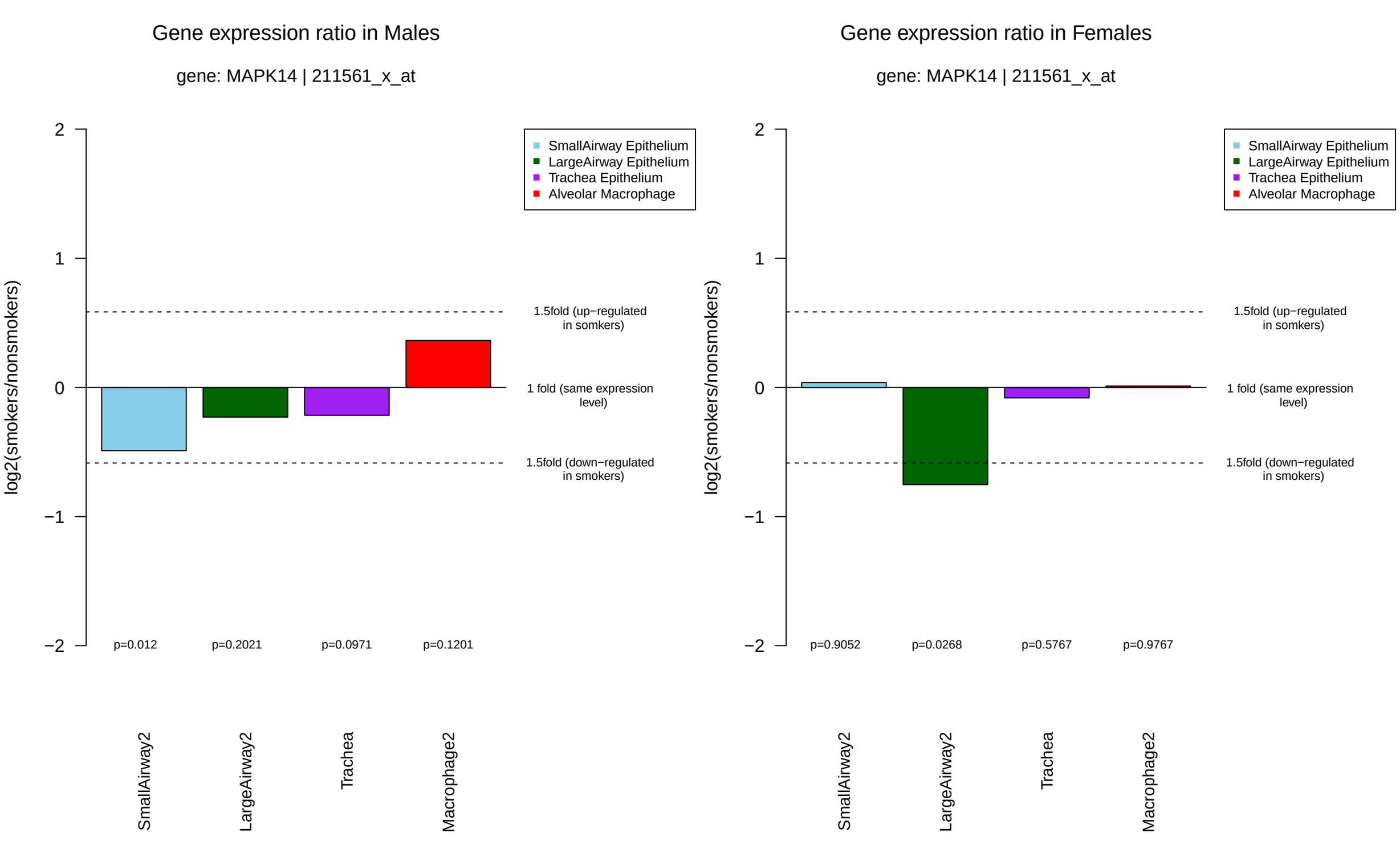
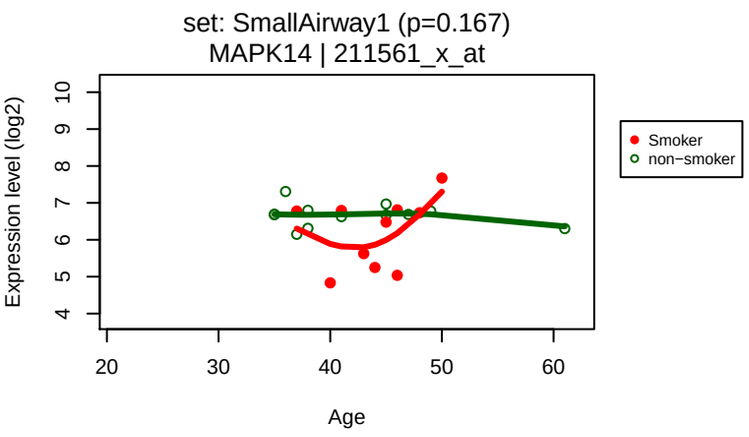
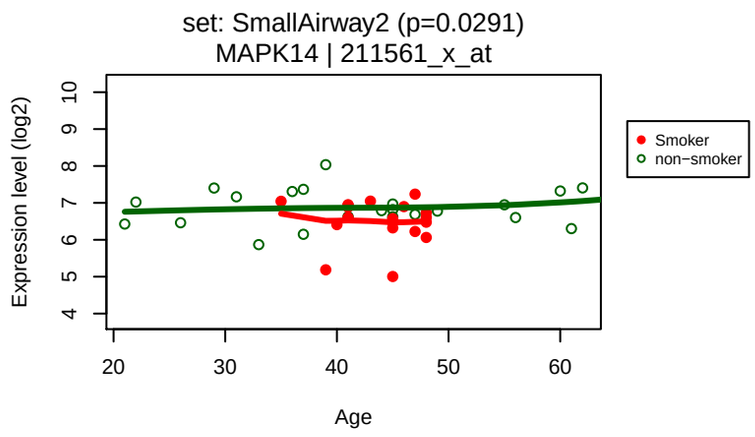
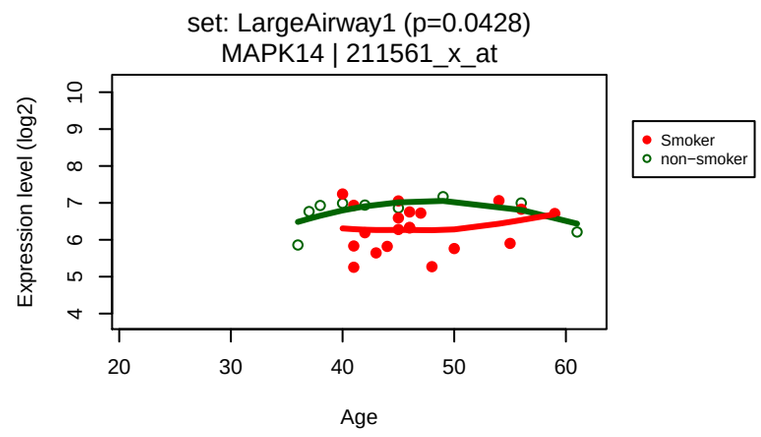

Gene expression of smokers and nonsmokers
gene: MAPK14 | 211561_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.167
p=0.0291
p=0.0137
p=0.0428
p=0.0056
p=0.0524
p=0.2607
p=0.2168
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

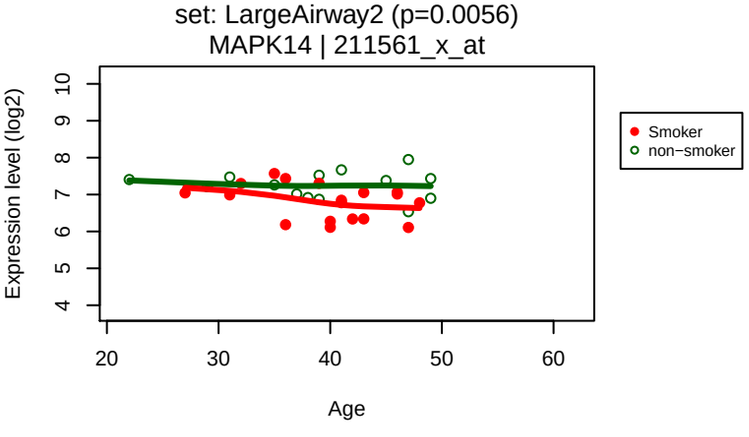

Gene expression ratio (smokers/nonsmokers)
gene: MAPK14 | 211561_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.167
p=0.0291
p=0.0137
p=0.0428
p=0.0056
p=0.0524
p=0.2607
p=0.2168
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

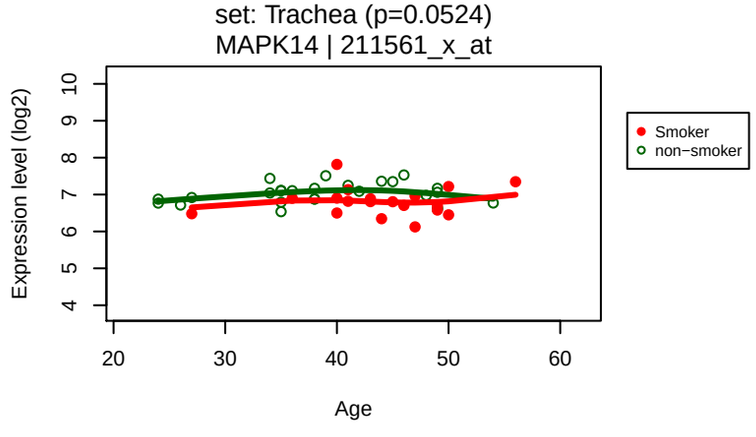

Gene expression ratio in Males
gene: MAPK14 | 211561_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.012
p=0.2021
p=0.0971
p=0.1201
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK14 | 211561_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9052
p=0.0268
p=0.5767
p=0.9767
SmallAirway2
LargeAirway2
Trachea
Macrophage2

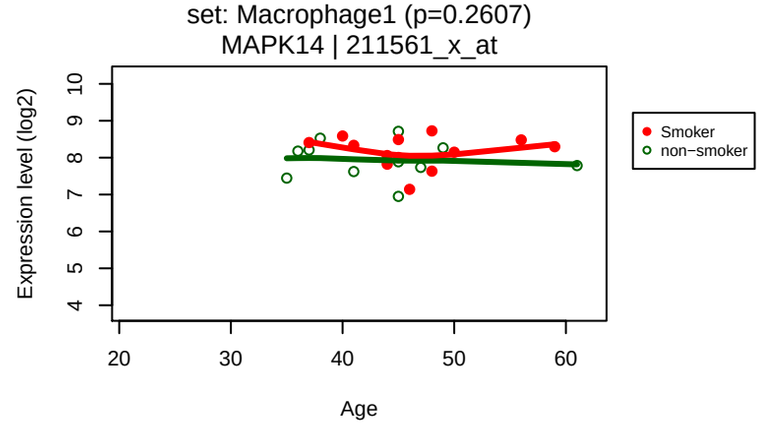

set: SmallAirway1 (p=0.167)
MAPK14 | 211561_x_at
Expression level (log2)
Age
Smoker
non−smoker

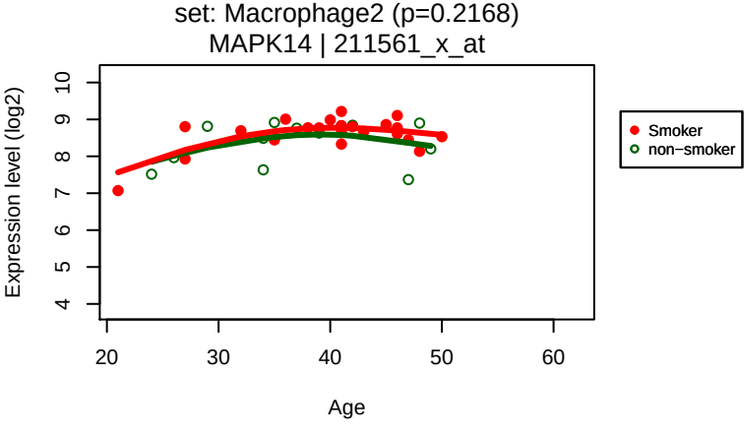

set: SmallAirway2 (p=0.0291)
MAPK14 | 211561_x_at
Expression level (log2)
Age
Smoker
non−smoker

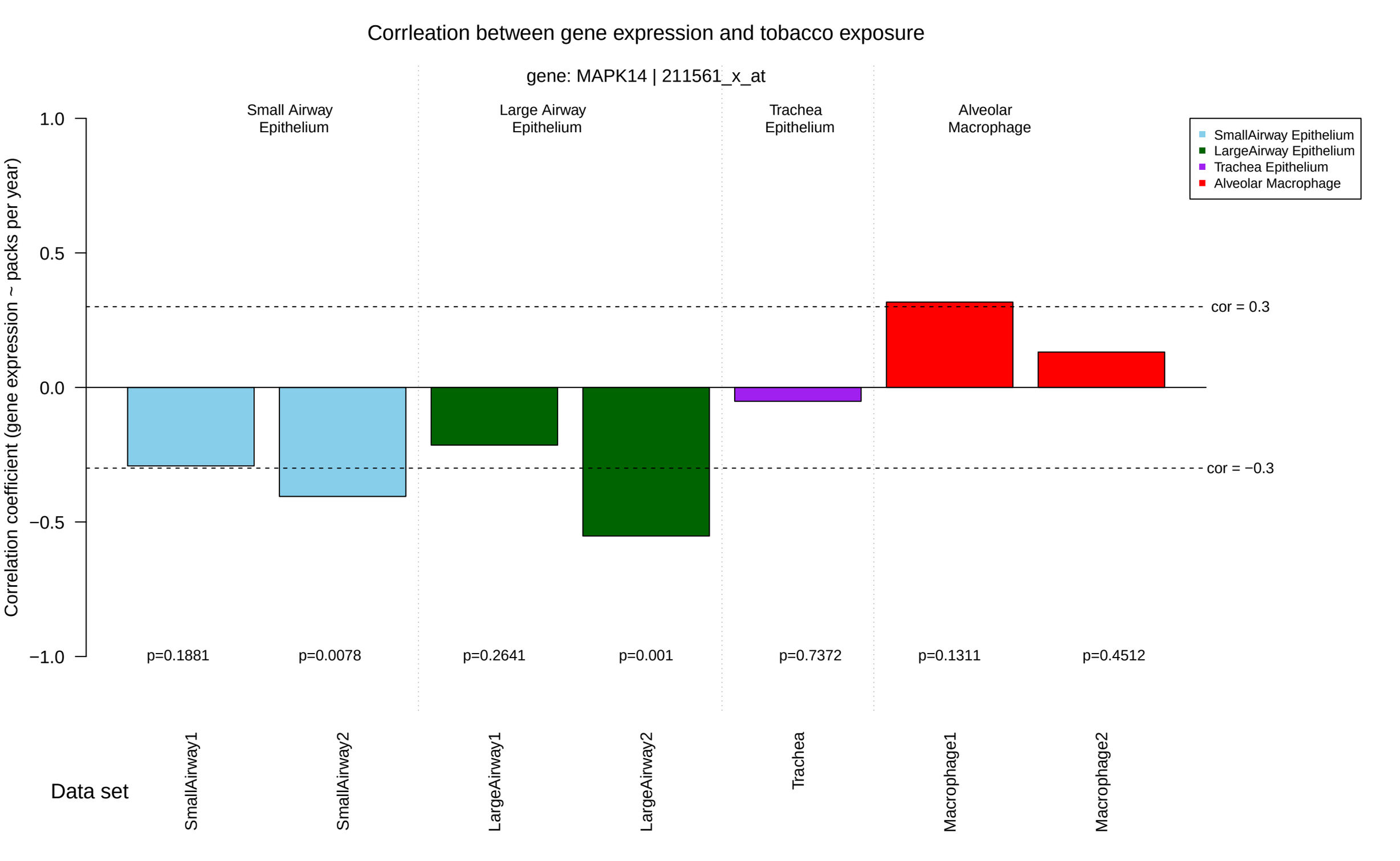

set: LargeAirway1 (p=0.0428)
MAPK14 | 211561_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0056)
MAPK14 | 211561_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0524)
MAPK14 | 211561_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2607)
MAPK14 | 211561_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2168)
MAPK14 | 211561_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK14 | 211561_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1881
p=0.0078
p=0.2641
p=0.001
p=0.7372
p=0.1311
p=0.4512
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2